

1 Choosing optimal lambda

Starting with our error bound, $\|f - q^*\|_\infty \lesssim \lambda^{-1}[e^{-cW \min\{\delta, \lambda\}} + W^{-1}e^{-\pi^2/\lambda}]$, we see the balance controlled by λ . With the other parameters fixed, increasing λ reduces the resolution and boundary bounds, suggesting that it should be pushed as large as possible. The aliasing term works in the opposite direction: increasing λ makes its bound worse.

We therefore set an aliasing budget e_{tol} and choose $\lambda^* = \max\{\lambda : E_{\text{alias}}(\lambda) \leq e_{\text{tol}}\}$ over admissible values, pushing λ as high as possible within that budget. This uses the predictable Fourier decay of the aliasing term to select λ without optimizing every error contribution at once. In practice, amplitude and fixed constants are absorbed into e_{tol} , with machine epsilon as a starting value.

Following this strategy, the general rule needs only the activation and tolerance and gives a starting value across widths and target functions. When a rough target frequency is available, the more specific rule also accounts for width.

To see what λ controls, consider $\tanh(\gamma(x - c))$. Its derivative is a bump of width about $1/\gamma$, so at center spacing $h = 2/N$ it spans about $1/(\gamma h) = 1/\lambda$ grid spacings. Increasing λ narrows these bumps and can leave dips between neighbors; decreasing λ increases their overlap and smooths those dips.

The unwanted wiggles between centers appear as extra Fourier components. Equal spacing ties the desired response to repetitions every $2\pi/h$, whose amplitudes relative to the desired component are determined by the kernel's Fourier transform.

We formalize the previous intuition with the following theorem:

Theorem (Alias bound). Under the representation theorem's assumptions, the boundary-corrected tanh construction has, for a unit-amplitude Fourier mode of angular frequency ω and $\theta = h|\omega| < \pi$,

$$E_{\text{alias}} \leq \mathcal{R}_{K,r}(\lambda, \theta) := \frac{2 \left[\left(\frac{\theta}{2\pi - \theta} \right)^r \widehat{K}\left(\frac{2\pi - \theta}{\lambda}\right) + \left(\frac{\theta}{2\pi + \theta} \right)^r \widehat{K}\left(\frac{2\pi + \theta}{\lambda}\right) \right]}{(1 - \rho_K) \widehat{K}(\theta/\lambda)}, \quad (1)$$

where $r = 1$, $K = \tanh'$, and $\rho_K = \widehat{K}((4\pi - \theta)/\lambda) / \widehat{K}((2\pi - \theta)/\lambda) < 1$.

For an activation ψ , $\widehat{K}$ is the Fourier transform of its localized derivative $K = \psi^{(r)}$. The numerator measures the first alias pair relative to the desired response in the denominator, while $1/(1 - \rho_K)$ bounds the remaining pairs; the proof is in Appendix A.

General rule. At fixed target frequencies, larger widths make θ small; retaining the rapid Fourier decay and absorbing slower prefactors into the tolerance gives the practical estimate

$$E_{\text{alias}}^{\text{basic}}(\lambda) := \frac{\widehat{K}(2\pi/\lambda)}{\widehat{K}(0)}. \quad (2)$$

For fp64 tanh, taking $e_{\text{tol}} = 2^{-52}$ gives $\lambda \approx 0.25$ as a practical default across admissible widths and analytic targets (Appendix A).

More specific rule. Retain the target-frequency shifts by choosing a rough angular frequency scale $\bar{\omega}$, such as 2π divided by a typical wavelength, and setting $\bar{\theta} = h\bar{\omega} < \pi$. Solve

$$\frac{\left(\frac{\bar{\theta}}{2\pi - \bar{\theta}} \right)^r \widehat{K}\left(\frac{2\pi - \bar{\theta}}{\lambda}\right) + \left(\frac{\bar{\theta}}{2\pi + \bar{\theta}} \right)^r \widehat{K}\left(\frac{2\pi + \bar{\theta}}{\lambda}\right)}{\widehat{K}(\bar{\theta}/\lambda)} = e_{\text{tol}} \quad (3)$$

for λ by bisection, then set $\gamma = \lambda/h$.